Ethan Xie
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EDUCATION

Yale University, New Haven, CT — Expected Jun 2030
B.S. Computer Science and Economics
Tesoro High School, Las Flores, CA — May 2026
GPA: 4.96/5.00 weighted academic; 4.00 unweighted

AWARDS AND HONORS

- California Seal of Civic Engagement (May 2026)
- President's Volunteer Service Award: Gold (August 2022, 2023, 2024)
- Tesoro Golf Coach's Award (May 2024)
- Tesoro Torch - AP Lang, AP Chemistry, and AP CSA Medal; Science Department Plaque (2023, 2024, 2025, 2026)
- Scored 5 in AP Calculus BC, Physics C Mechanics, Chemistry, Biology, English Literature, Chinese Language, Macroeconomics, American Government, Computer Science A, United States History

WORK EXPERIENCE

NGEX Webworks, *Co-Founder and Head of Technology*, Los Angeles, CA — Sep 2024 – May 2026
- Built AI-enabled websites for small businesses, improving customer engagement and online conversion
- Generated over $5,000 in revenue through client projects

Treasure Hacks, *Software Engineering and Community Management Intern*, San Diego, CA — Jan 2023 – May 2026
- Organized hackathons with over 250 participants and $80k+ in prizes
- Attended Cause San Diego with 50+ industry professional and created relationships based on passion for social impact
- Developed presentation game to teach presentation skills using MongoDB and sockets
- Built website for and hosted 8 hour in person skills workshop with 60+ attendees

City of Laguna Niguel, *Recreation Leader*, Laguna Niguel, CA — Oct 2024 – May 2026
- Served seniors in the city of Laguna Niguel, providing refreshments, helping with technology, and conversating
- Managed events hosted by third parties such as weddings, birthday parties, religious group meetings, and conferences

LEADERSHIP EXPERIENCE

Tesoro IMPACT, *Executive Lead* — Aug 2022 – May 2026
- Hosted and gave speeches at freshman orientation and homerooms for two years with classes of 400+ freshmen
- Organized multiple school-wide contests and food drives across the years to promote community, benefiting food pantries like South Coast Outreach, donating over 1000+ cans

Tesoro Computer Science Club, *President* — Aug 2022 – May 2026
- Organized meetings and outreach events with a focus on promoting computer science education and engagement
- Assisted 20+ members in developing projects and passion for computer science

SELECTED PROJECTS

Transformers - *Python, Pytorch* — *https://github.com/Anonymouseyy/transformers*
- Implemented a decoder-only transformer architecture from scratch in PyTorch without relying on high-level libraries
- Learned the in-depth functions of multi-head self-attention, layer normalization, and residual connections
- Trained a decoder-only transformer on character-level/text datasets and analyzed effects of model size and hyperparameters

Autograd Engine - *Python, Calculus* — *https://github.com/Anonymouseyy/micrograd*
- Developed a reverse-mode automatic differentiation engine from scratch in Python based on Karpathy's Micrograd
- Implemented computational graphs, gradient propagation, and backpropagation for neural network training
- Used the engine to train a convolutional neural network on MNIST

FELLOWSHIPS

Yale AI Alignment — Jul 2026 – Aug 2026
- Studied technical and conceptual problems in AI alignment, including misuse, outer alignment, and mesa-optimization

Yale Effective Altruism — Jul 2026 – Aug 2026
- Explored research on global health, AI safety, longtermism, and cost-effectiveness through weekly discussions and readings

ADDITIONAL INFORMATION

Skills: Python, PyTorch, scikit-learn, Pandas, NumPy, Next.js, Git, Chinese
Interests: Golf, Dostoevsky, Tolstoy, Counter Strike, Guitar